

Brain-Computer Interface

Project Overview

- **Duration:** 5 Months
 - **Project Mentors:** Charan, Dhanya
 - **Team Members:** Hadi Hasan, Aryan, Hithinsai, Chitropal
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1. Introduction

1.1 Project Statement

This project aims to design and implement a low-cost Brain-Computer Interface (BCI) system capable of detecting eye blinking (EOG) and forearm muscle movement (EMG) and converting them into digital control signals using an embedded system.

1.2 Background Survey

Brain-Computer Interfaces (BCIs) enable communication between humans and machines through physiological signals. EOG captures eye movements in the low-frequency range, while EMG captures muscle activity in higher frequency ranges. Compared to EEG-based systems, EOG and EMG provide a simpler and cost-effective alternative for real-time applications.

2. Methodology

2.1 Techniques & Approach

The system uses differential signal acquisition through electrodes, followed by amplification using an instrumentation amplifier. Bandpass filters isolate the required frequency ranges for EOG and EMG signals. The processed signals are then fed into an ESP32 microcontroller for detection.

2.2 System Architecture & Workflow

The workflow consists of:

- Signal acquisition using electrodes
- Amplification using instrumentation amplifier
- Bandpass filtering
- Signal processing using ESP32
- Output indication using LED

2.3 Pseudocode

ARDUINO CODE

START

SET SIGNAL_PIN = 34

SET WINDOW_SIZE = 10

INITIALIZE array readings[WINDOW_SIZE] = {0}

SET idx = 0

SET sum = 0

SET thresholdBlink = 250

SET thresholdMuscle = 450

SET lastTrigger = 0

SET cooldown = 500 ms

FUNCTION movingAverage(newValue):

 SUBTRACT readings[idx] from sum

 STORE newValue in readings[idx]

 ADD newValue to sum

 UPDATE idx = (idx + 1) mod WINDOW_SIZE

 RETURN sum / WINDOW_SIZE

END FUNCTION

Initialize serial communication

LOOP forever:

 READ raw signal from SIGNAL_PIN → raw

 CALL movingAverage(raw) → smooth

 GET current time using millis()

 IF (current time - lastTrigger > cooldown) THEN

 IF (smooth > thresholdBlink AND smooth < thresholdMuscle) THEN

 PRINT "BLINK"

 UPDATE lastTrigger = current time

 ELSE IF (smooth > thresholdMuscle) THEN

 PRINT "MUSCLE"

 UPDATE lastTrigger = current time

 END IF

 END IF

 WAIT for 5 milliseconds

END LOOP

END

PYTHON CODE

START

IMPORT serial communication library

IMPORT keyboard control library

INITIALIZE serial connection on COM3 with baud rate 115200

PRINT "BCI Control Started..."

```

LOOP forever:

    TRY:

        READ incoming data from serial → line
        DECODE line to text
        REMOVE extra spaces/newlines

        IF line is not empty THEN
            PRINT "Detected:", line

        IF line == "BLINK" THEN
            PRESS "space" key
            PRINT "Jump"

        ELSE IF line == "MUSCLE" THEN
            PRESS "down" key
            PRINT "Crouch"

    CATCH any error:
        PRINT error message

END LOOP

END

```

2.4 Tools & Technologies Used

Hardware:

- Instrumentation Amplifier (AD620)
- Operational Amplifiers
- Surface Electrodes
- ESP32 Microcontroller
- Oscilloscope
- $\pm 9V$ Power Supply

Software:

- Arduino IDE
- LTSpice

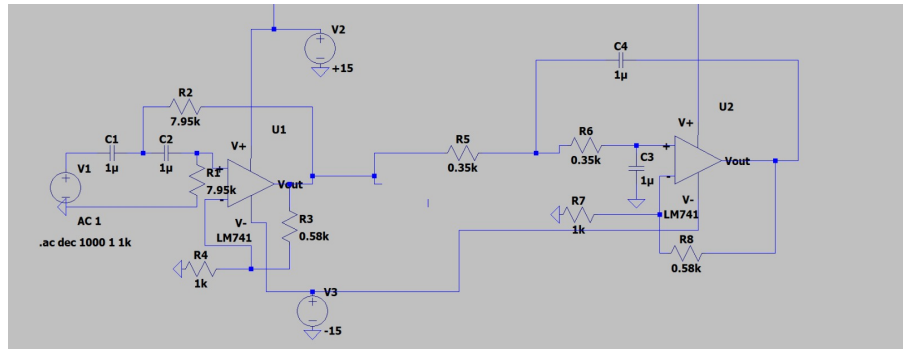


Figure 1: Bandpass Filter

3. Results & Analysis

The system successfully detected both EOG and EMG signals. Eye blink signals appeared as slow, high-amplitude spikes, while muscle activity appeared as high-frequency bursts. The ESP32 successfully processed the signals and triggered LED outputs in real-time.

Some noise was observed due to environmental interference, but the system performed reliably under controlled conditions.

4. Challenges & Learnings

4.1 Obstacles Faced

- Low amplitude of bio-signals
- Noise interference (especially 50 Hz)
- Electrode placement difficulties
- Signal instability due to movement

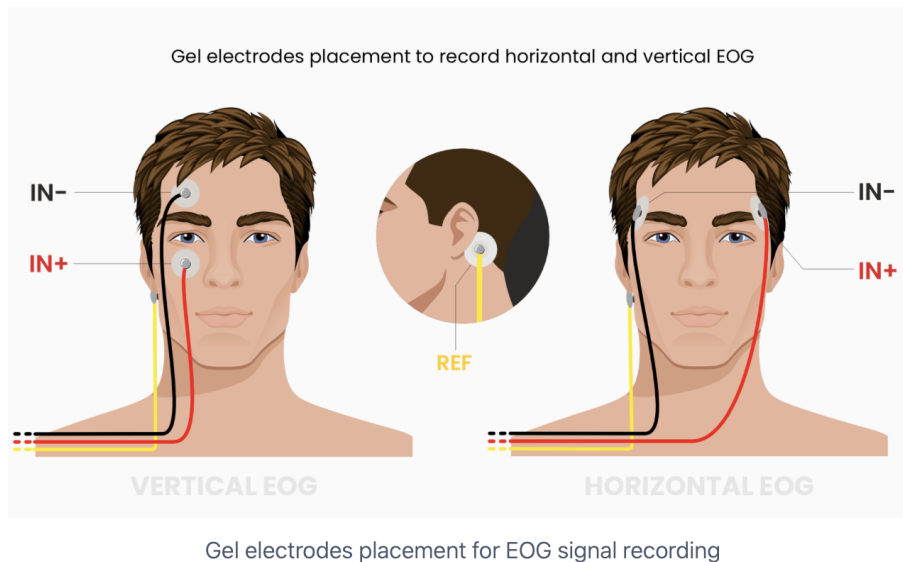


Figure 2: EOG Placement

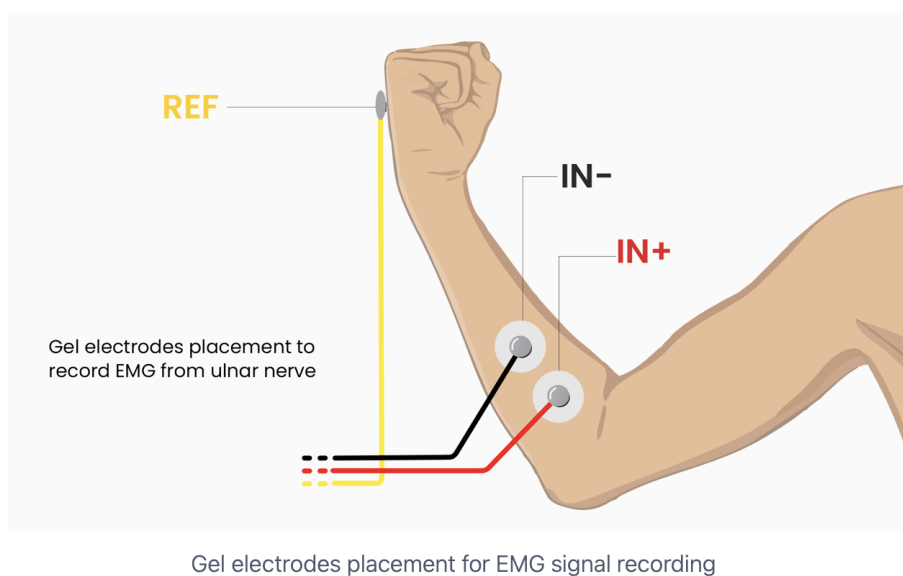


Figure 3: EMG Placement

4.2 Solutions & Insights

- Used instrumentation amplifier for signal gain
- Applied bandpass filtering
- Optimized electrode placement
- Leveraged differential measurement to reduce noise

5. Conclusion & Applications

5.1 Final Takeaways

A functional low-cost BCI system was successfully implemented, demonstrating real-time detection of bio-signals and their conversion into control outputs.

5.2 Real-World Applications

- Assistive technologies
 - Human-machine interaction
 - Gesture-based control systems
 - Prosthetic control
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6. Future Work & Enhancements

- Add 50 Hz notch filter
 - Implement machine learning classification
 - Improve signal accuracy
 - Enable wireless communication
 - Control external devices (robot, wheelchair)
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7. References

1. J. G. Webster, *Medical Instrumentation*
2. IEEE Papers on EMG Signal Processing
3. ESP32 Technical Reference Manual
4. Analog Devices Datasheets (AD620)
5. LTSpice Documentation